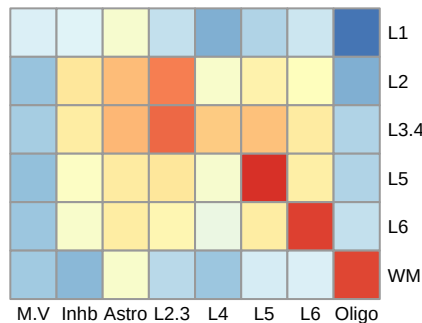
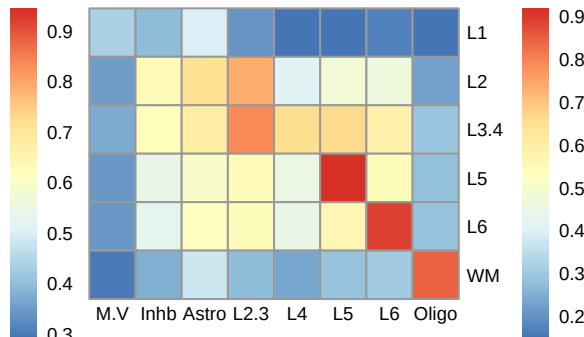


t statistic correlation between L-R annotations

NTC.MDD F



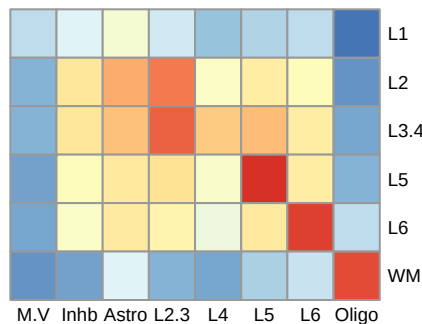
NTC.MDD M



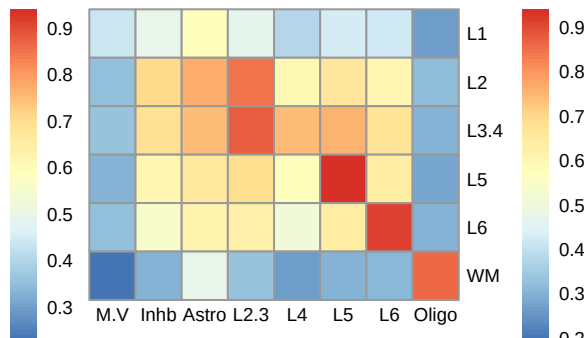
L5, L6, and WM/Oligo have extremely similar L-R results from the two annotations, consistent with the high degree of spot-level overlap between each pair of annotations.

There is a low degree of overlap between PRECAST L1 L-R results and Seurat label Micro.Vasc or Astro L-R results, despite some overlap in spot-level annotations.

NTC.BPD F

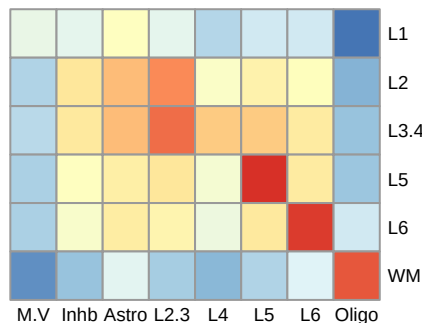


NTC.BPD M

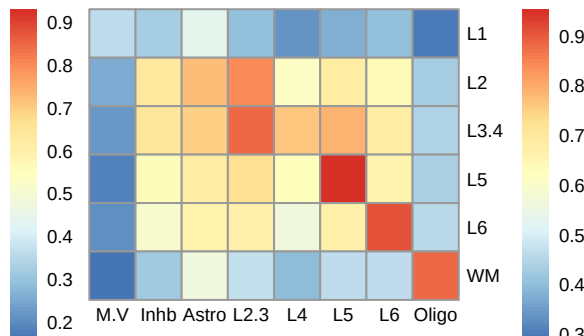


There is no clean way to look at the supra-granular comparisons between the PRECAST and Seurat label annotations. Both from the biological justification of overlapping importance of each of L2 vs L2.3 and L3.4 vs L2.3 and L3.4 vs L4, as well as from looking at the correlation of the t-statistics of the L-R results.

MDD.BPD F

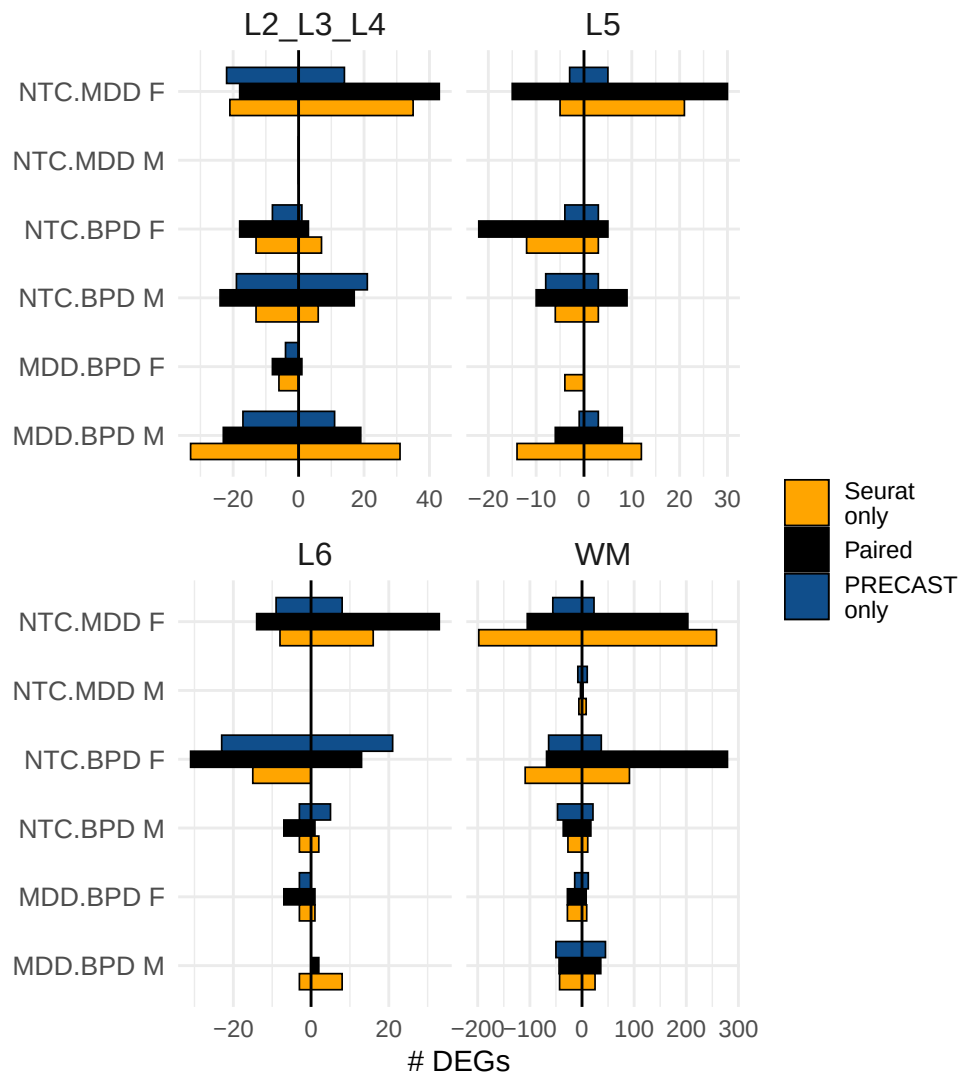


MDD.BPD M

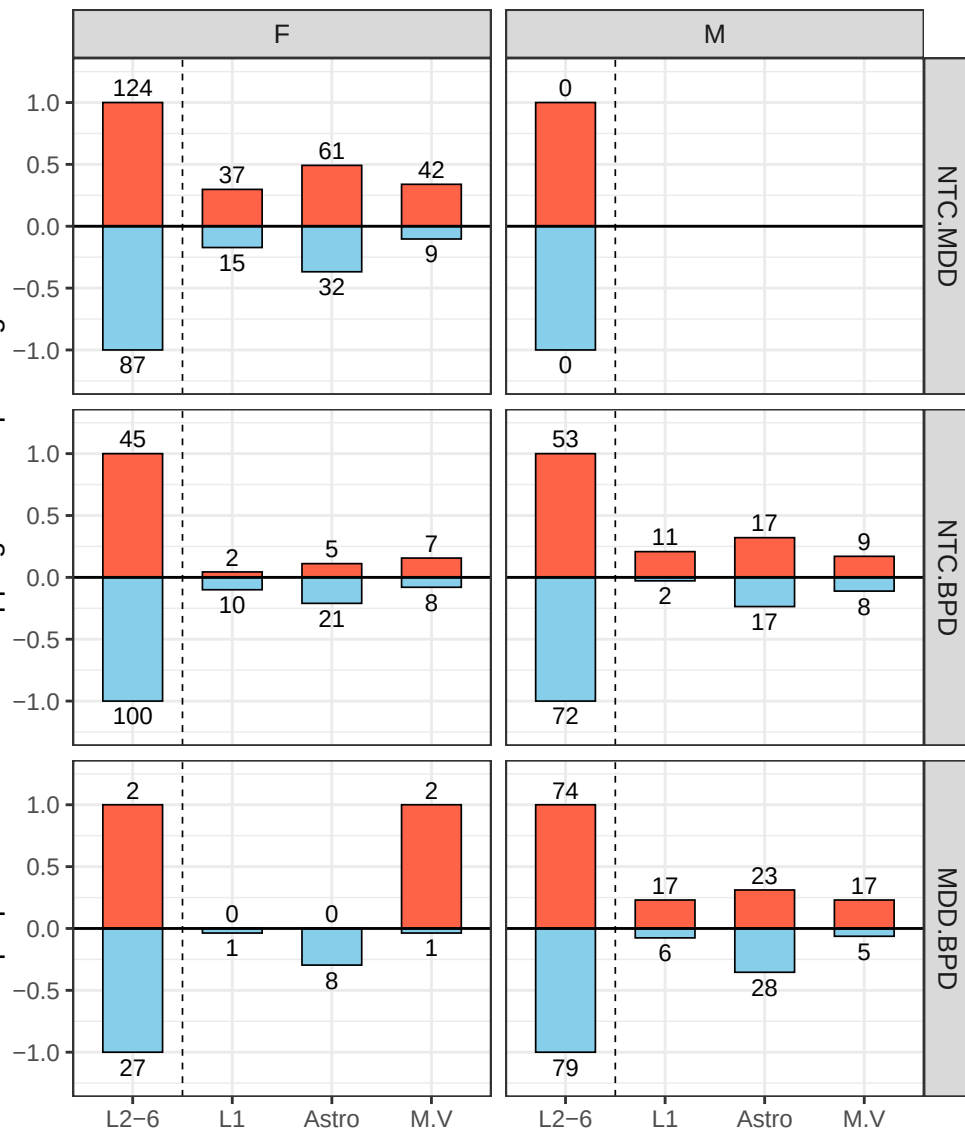


Altogether, we will look at a combined group of L2,L3,L4 for L-R DEG consistency, as well as the 3 separate groups of L5, L6, and WM.

Number of paired L-R DEGs across matching groups



prop. of L2-6 DEGs overlapping with un-paired glial DEGs



This plot examines the overlap between any neuronal (L2-L6) DEGs from either annotation and the glial DEGs of L1 (PRECAST), Micro.Vasc (Seurat), and Astro (Seurat).

For each dx*sex comparison (facets), the y-axis indicates the proportion of L2-6 DEGs that were also DEGs in the x-axis groups. Positive/ negative values indicate whether DEGs were increased/ decreased, respectively. To help orient, the vertical dashed line separates the gene set being used in the denominator (on the left; L2-6 DEGs), versus the glial DEGs being used in the numerator (on the right).

The total number of DEGs in each bar is given by the text value, to help discern when a large proportion of L2-6 DEGs being identified as glial DEGs is due to low overall number of L2-6 DEGs. These values are also direction (increased/ decreased) specific.

NTC.MDD F

Decreased

			L1	L2	L3.4	L5	L6	WM
<i>PRECAST DEG</i>			113	31	19	18	23	161
<i>PRECAST only</i>				17	5	3	9	56
<i>Paired DEG</i>				18		15	14	105
<i>Seurat only</i>				11	10	5	8	198
<i>Seurat DEG</i>	6	291	65	26	22	20	22	303
	Inhb	M.V	Astro	L2.3	L4	L5	L6	Oligo

Increased

			L1	L2	L3.4	L5	L6	WM
<i>PRECAST DEG</i>			152	42	40	35	41	226
<i>PRECAST only</i>				8	8	5	8	23
<i>Paired DEG</i>				43		30	33	203
<i>Seurat only</i>				19	20	21	16	258
<i>Seurat DEG</i>	20	337	104	62	42	51	49	461
	Inhb	M.V	Astro	L2.3	L4	L5	L6	Oligo

NTC.MDD M

Decreased

			L1	L2	L3.4	L5	L6	WM
<i>PRECAST DEG</i>			6	0	0	0	0	11
<i>PRECAST only</i>				0	0	0	0	8
<i>Paired DEG</i>				0		0	0	3
<i>Seurat only</i>				0	0	0	0	6
<i>Seurat DEG</i>	0	92	0	0	0	0	0	9
	Inhb	M.V	Astro	L2.3	L4	L5	L6	Oligo

Increased

			L1	L2	L3.4	L5	L6	WM
<i>PRECAST DEG</i>			11	0	0	0	0	12
<i>PRECAST only</i>				0	0	0	0	10
<i>Paired DEG</i>				0		0	0	2
<i>Seurat only</i>				0	0	0	0	8
<i>Seurat DEG</i>	0	81	0	0	0	0	0	10
	Inhb	M.V	Astro	L2.3	L4	L5	L6	Oligo

NTC.BPD F

Decreased

			L1	L2	L3.4	L5	L6	WM
<i>PRECAST DEG</i>			42	19	10	26	54	132
<i>PRECAST only</i>				6	2	4	23	64
<i>Paired DEG</i>				18		22	31	68
<i>Seurat only</i>				7	7	12	15	109
<i>Seurat DEG</i>	6	83	25	25	14	34	46	177
	Inhb	M.V	Astro	L2.3	L4	L5	L6	Oligo

Increased

			L1	L2	L3.4	L5	L6	WM
<i>PRECAST DEG</i>			38	4	2	8	34	316
<i>PRECAST only</i>				1	0	3	21	37
<i>Paired DEG</i>				3		5	13	279
<i>Seurat only</i>				4	3	3	0	91
<i>Seurat DEG</i>	5	64	8	7	3	8	13	370
	Inhb	M.V	Astro	L2.3	L4	L5	L6	Oligo

NTC.BPD M

Decreased

			L1	L2	L3.4	L5	L6	WM
<i>PRECAST DEG</i>			39	29	25	18	10	83
<i>PRECAST only</i>				13	6	8	3	47
<i>Paired DEG</i>				24		10	7	36
<i>Seurat only</i>				10	3	6	3	27
<i>Seurat DEG</i>	2	123	23	34	3	16	10	63
	Inhb	M.V	Astro	L2.3	L4	L5	L6	Oligo

Increased

			L1	L2	L3.4	L5	L6	WM
<i>PRECAST DEG</i>			71	28	17	12	6	38
<i>PRECAST only</i>				15	6	3	5	21
<i>Paired DEG</i>				17		9	1	17
<i>Seurat only</i>				4	2	3	2	11
<i>Seurat DEG</i>	7	147	23	19	7	12	3	28
	Inhb	M.V	Astro	L2.3	L4	L5	L6	Oligo

MDD.BPD F

Decreased

			L1	L2	L3.4	L5	L6	WM
<i>PRECAST DEG</i>			23	11	2	0	10	42
<i>PRECAST only</i>				4	0	0	3	14
<i>Paired DEG</i>				8		0	7	28
<i>Seurat only</i>				4	2	4	3	28
<i>Seurat DEG</i>	0	128	17	12	3	4	10	56
	Inhb	M.V	Astro	L2.3	L4	L5	L6	Oligo

Increased

			L1	L2	L3.4	L5	L6	WM
<i>PRECAST DEG</i>			38	0	1	0	1	20
<i>PRECAST only</i>				0	0	0	0	12
<i>Paired DEG</i>				1		0	1	8
<i>Seurat only</i>				0	0	0	1	9
<i>Seurat DEG</i>	0	182	1	1	0	0	2	17
	Inhb	M.V	Astro	L2.3	L4	L5	L6	Oligo

MDD.BPD M

Decreased

			L1	L2	L3.4	L5	L6	WM
<i>PRECAST DEG</i>			103	16	27	7	0	94
<i>PRECAST only</i>				9	8	1	0	50
<i>Paired DEG</i>				23		6	0	44
<i>Seurat only</i>				15	20	14	3	43
<i>Seurat DEG</i>	5	151	59	35	32	20	3	87
	Inhb	M.V	Astro	L2.3	L4	L5	L6	Oligo

Increased

			L1	L2	L3.4	L5	L6	WM
<i>PRECAST DEG</i>			140	15	21	11	2	81
<i>PRECAST only</i>				4	7	3	0	45
<i>Paired DEG</i>				19		8	2	36
<i>Seurat only</i>				6	26	12	8	25
<i>Seurat DEG</i>	10	208	44	22	34	20	10	61
	Inhb	M.V	Astro	L2.3	L4	L5	L6	Oligo